



H2S

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : Team Anant

Team Leader Name : Adarsh Jayprakash Mishra

Problem Statement : Forecasting and/or Nowcasting of Solar Flares using combined Soft and Hard X-ray data from Aditya-L1

Team Members

Team Leader:

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Team Member-1:

Name: Deepanshi Rathee
College: Govt. P.G. COLLEGE FOR
WOMEN, ROHTAK

Team Member-2:

Name: Hindavi Rajendra Kudu
College: Sonopant Dandekar Shikshan
Madali, Palghar

Opportunity should be able to explain the following:

How different is it from any of the other existing ideas?

- Most solutions analyse a single X-ray source, while our approach combines both SoLEXS (Soft X-ray) and HEL1OS (Hard X-ray) observations.
- Instead of generating separate outputs, our system creates a unified flare catalogue from multiple instruments.
- The solution focuses on both flare detection and future nowcasting using the same integrated pipeline.
- A functional prototype has already been developed and tested on real Aditya-L1 datasets.

How will it be able to solve the problem?

- By continuously analysing Soft and Hard X-ray observations together.
- By detecting abnormal flare activity through statistical analysis and peak identification.
- By correlating events across multiple instruments to improve reliability.
- By generating early indicators that can support solar flare nowcasting.

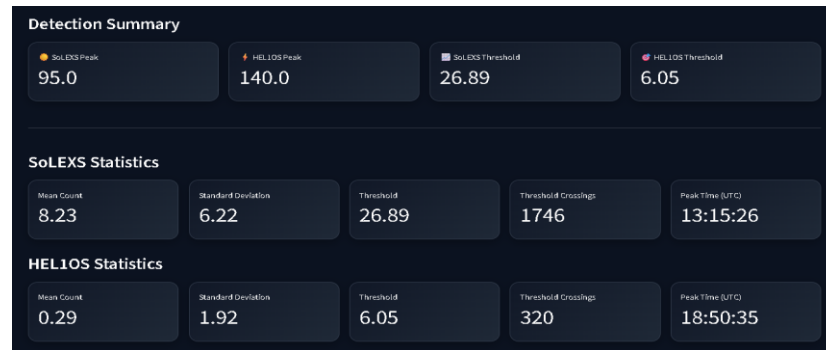
USP of the proposed solution

- Combined Soft X-ray + Hard X-ray analysis.
- Unified Solar Flare Catalogue generation.
- Real Aditya-L1 data integration.
- Interactive visualization dashboard.
- Scalable architecture for nowcasting and forecasting.

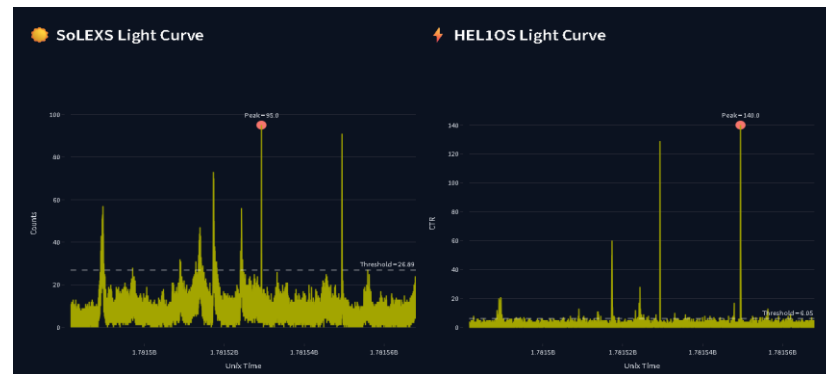
List of features offered by the solution

- Processes SoLEXS and HEL10S datasets
- Automatic solar flare detection
- Dynamic threshold calculation
- Peak identification and analysis
- Unified Solar Flare Catalogue generation
- Interactive visualization dashboard
- Future-ready nowcasting support
- Detection of low- and high-class flares
- High True Positive Rate and low False Alarm Rate for predictions.

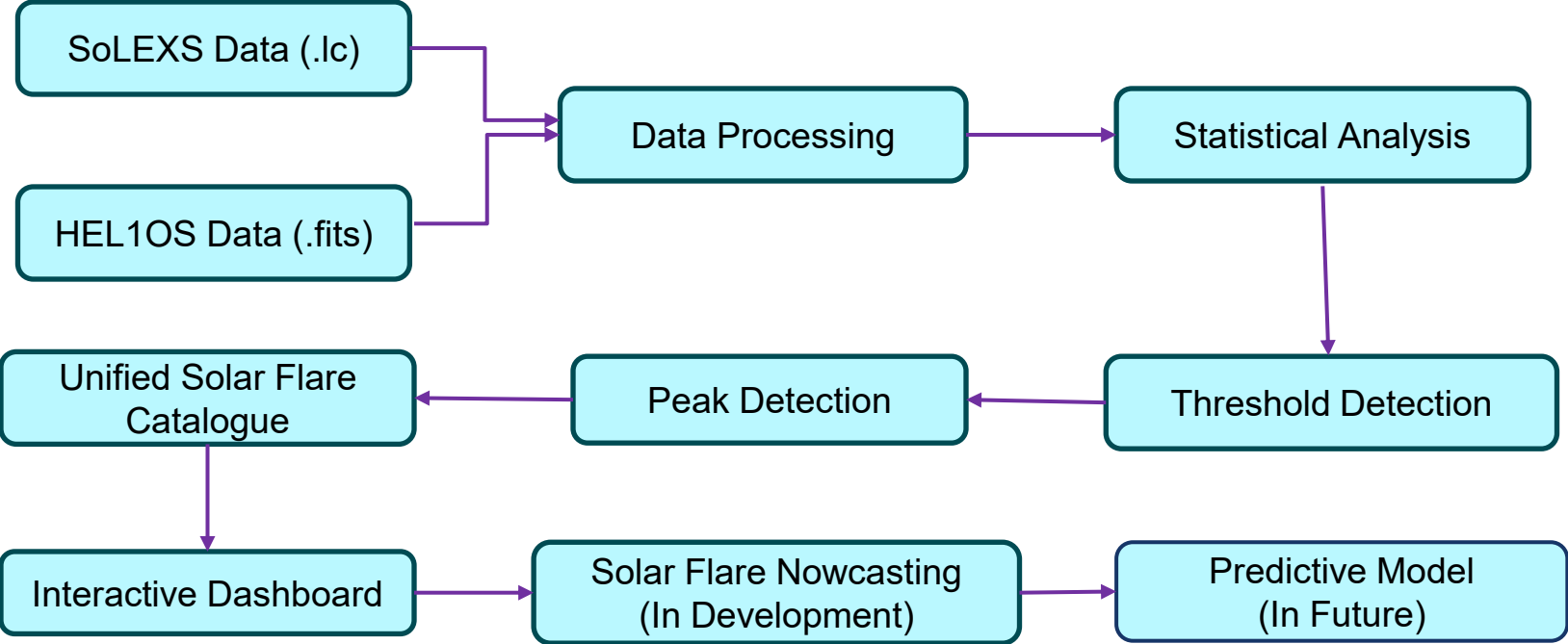
Detection Summary :



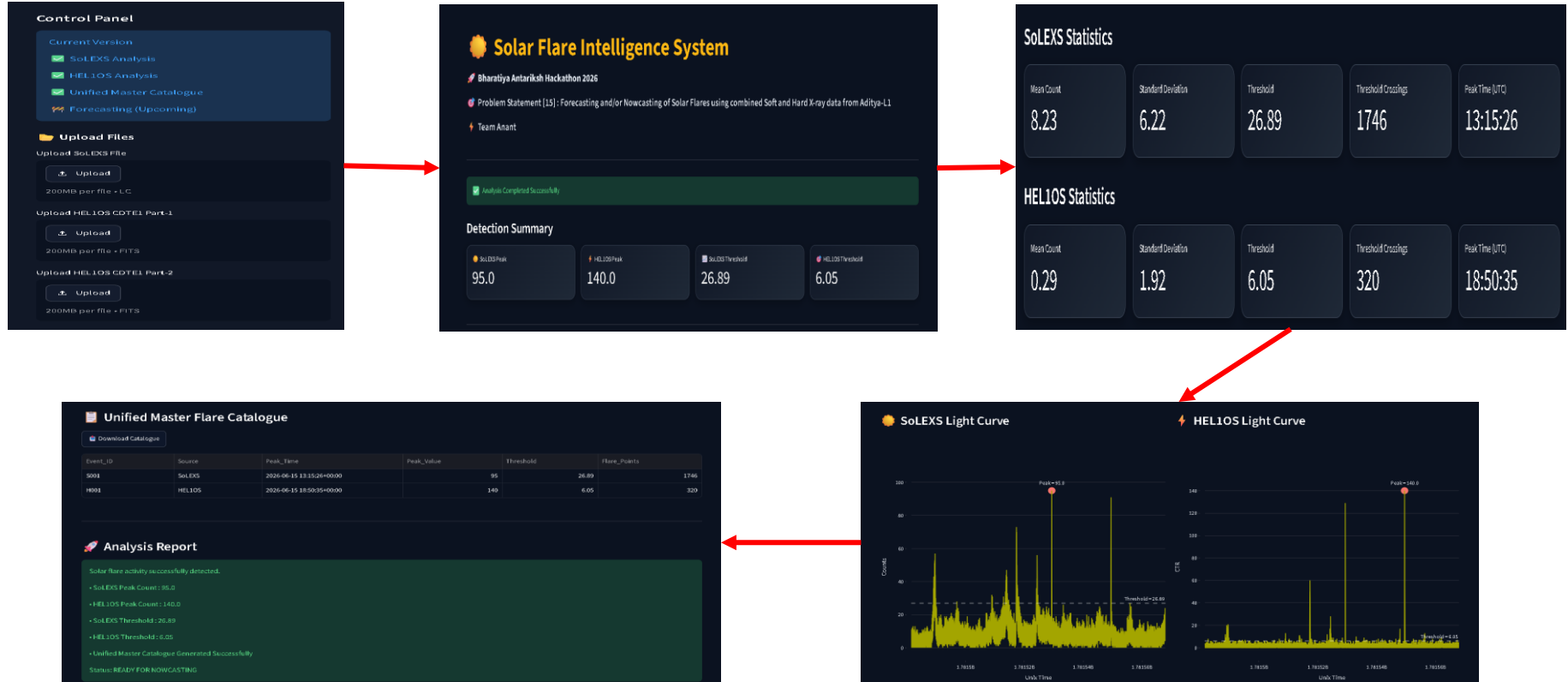
Light Curve :



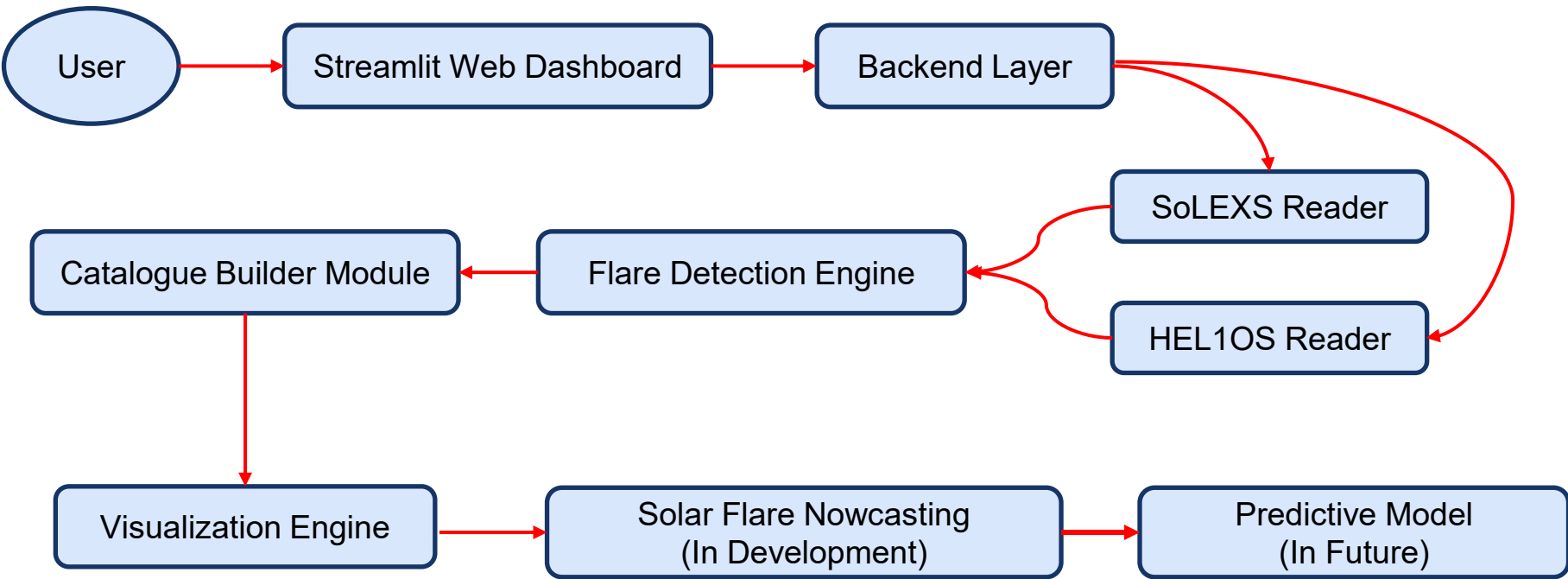
Process flow diagram or Use-case diagram :



Wireframes/Mock diagrams of the proposed solution (optional)



Architecture diagram of the proposed solution



Technologies to be used in the solution:

 Frontend → Streamlit

 Backend → Python

 Analytics → Pandas, NumPy

 Scientific Processing → Astropy

 Visualization → Plotly, Matplotlib

 Data Sources → Aditya-L1 SoLEXS, HEL1OS

 Version Control → Git, GitHub

 Deployment → Streamlit Cloud

Estimated implementation cost (optional):

Prototype Cost: ₹0

- Open-source Python ecosystem
- Free scientific libraries
- Publicly available Aditya-L1 datasets
- Free GitHub hosting
- Free Streamlit Cloud deployment

Estimated Cost (Prototype Phase): ₹0

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THANK YOU